Lab 6 – Code Coverage Analysis Report

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Application: `password_policy.py`
Test file: `CodeCoverageTest_AM039_application.py`

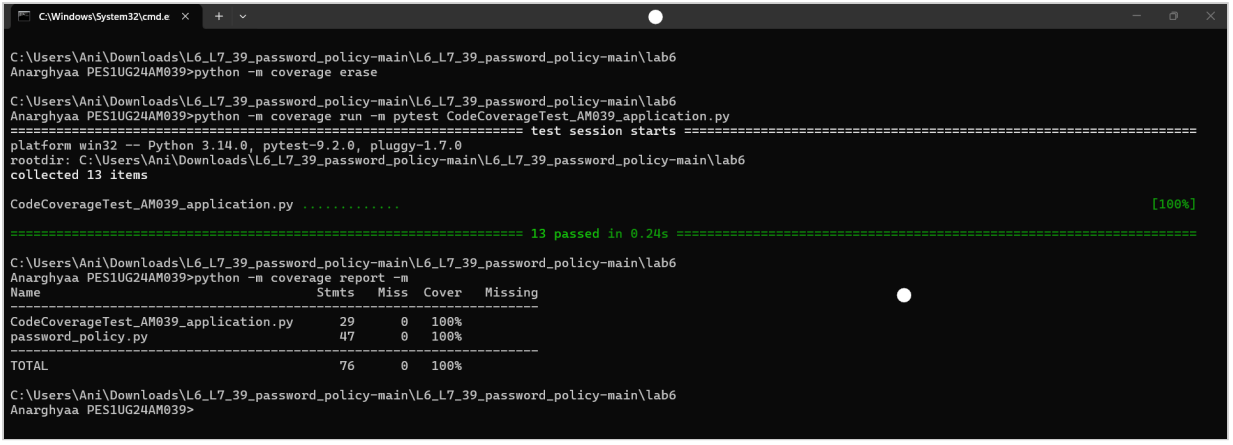
Commands used

```
python -m coverage run -m pytest CodeCoverageTest_AM039_application.py
python -m coverage report -m
```

Result

	Starter tests	After new tests
Tests	2	13 (all passed)
Statements in `password_policy.py`	47	47
Missed statements	21	0
Code coverage	**55%**	**100%**

Screenshot



Analysis

- The starter tests only used a valid password (`"Abcdef1!"`), so the lines that report problems, the `strong / weak / very weak` ratings and the whole `suggest_fix` function never ran.
- A test was added for each missed line:
  - **Too short:** `"Ab1!"`
  - **Too long, plus no uppercase, digit or special character:** `"a" * 65`
  - **Common password, plus no lowercase:** `"PASSWORD"`
  - **Strong:** `"Abcdefghijk1!@"` (14 characters)
  - **Weak:** `"Abcdefgh1"` (one issue)
  - **Very weak:** `"abc"` (four issues)
  - **`suggest_fix` :** one input for each of its five return values
- Every line of `password_policy.py` now executes at least once, so code coverage is **100%**, above the 90% target.
- Code coverage only shows that each line ran. It does not show that both outcomes of every `if` were tested, which is what branch coverage checks.